ICU Mortality Prediction using Machine Learning

Course: Machine Learning (CS4082)

Done by: Mashael Saeed, Sarah Elshiaty, Seifeldin Elshiaty

Instructor: Dr. Naila Marir

HEALTHCARE AI

MACHINE LEARNING

Problem Definition & Research Gap

HEALTHCARE CONTEXT

Problem Definition

ICU patients can deteriorate quickly, so early mortality-risk prediction is important for supporting timely clinical decisions.

Project Goal:

Build a machine learning model that predicts ICU mortality risk using structured clinical data and classifies patients into risk levels.

METHODOLOGICAL CHALLENGE

Research Gap

In healthcare prediction tasks, accuracy alone can be misleading, especially when the dataset is imbalanced and survival cases outnumber death cases. That's why healthcare requires **recall-focused evaluation.**

A model may achieve acceptable accuracy but still miss **high-risk patients**, which is dangerous in ICU settings.

OUR APPROACH

Our Focus

- Handle **class imbalance**
- Prioritize **recall** to reduce missed **high-risk patients**
- Use clinically meaningful features
- Support interpretability through feature importance and risk stratification

Project Positioning & Significance

PROJECT POSITIONING

A healthcare machine learning **decision-support system** for ICU mortality risk prediction.

Important note: The system **does not replace clinicians**. It supports early identification of **high-risk ICU patients** by providing mortality probability, predicted risk category, and interpretable clinical risk factors.

Why It Matters

ICU patients can deteriorate quickly, and ICU resources are limited. Early risk prediction can help prioritize patients who may need closer monitoring or urgent intervention.

Key Significance:

- Supports earlier recognition of critical patients
- Helps reduce missed high-risk cases
- Provides interpretable predictions using clinical features
- Demonstrates responsible AI use in healthcare

SDG Alignment

SDG 3: GOOD HEALTH AND WELL-BEING

Supports improved patient care by helping identify ICU patients at higher mortality risk.

SDG 9: INDUSTRY, INNOVATION AND INFRASTRUCTURE

Applies AI and machine learning to healthcare innovation and digital medical systems.

SAUDI VISION 2030

Supports digital transformation, healthcare innovation, AI-based decision support, and improved quality of life through smarter health services.

Dataset Overview & Data Preprocessing

DATASET OVERVIEW

ICU Clinical Data

The dataset contains **15,000 ICU patient records** used to predict mortality risk.

It includes **clinical and physiological features** collected at or near admission, describing the patient's health condition, such as:

- Vital signs and laboratory measurements
- Patient characteristics, such as age and medical conditions
- Clinical indicators related to patient status
- Target variable: **mortality_label**
  - **0 = survived**
  - **1 = died**

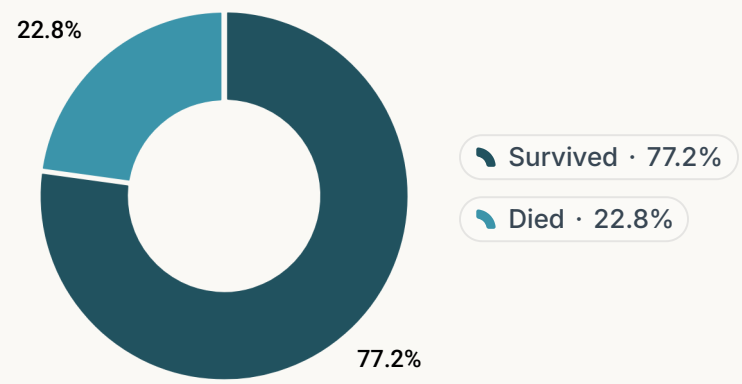
KEY DATA CHALLENGE

Class Imbalance

The dataset was **imbalanced**; survival cases significantly outnumbered death cases.

Imbalanced dataset:

- **Survived:** 11,582
- **Died:** 3,418



Why this matters: Model may become biased toward predicting survival, missing **high-risk mortality cases**.

PREPROCESSING PIPELINE

Data Transformation Steps

- Removed irrelevant and leakage features
- Handled missing values using imputation
- Checked outliers
- Encoded categorical and binary variables
- Scaled numerical features
- Applied train/test split (no leakage)

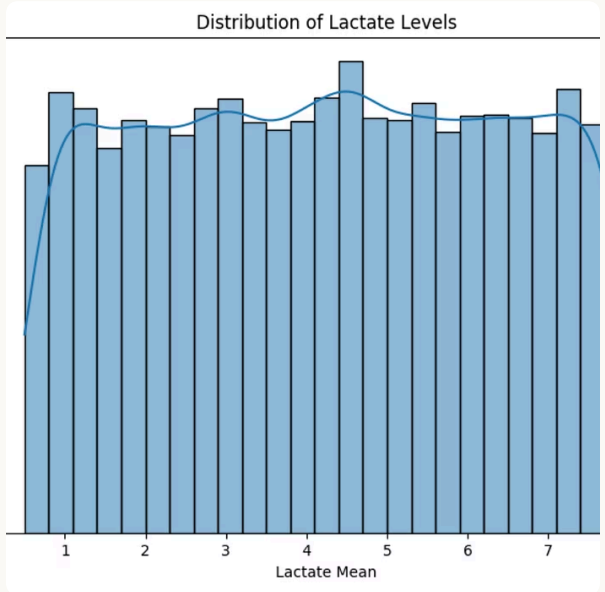
FEATURE REDUCTION

Reduced from 22 → 7 final features:

- ventilation_required
- vasopressor_used
- sepsis_flag
- apache_score
- sofa_score
- comorbidity_score
- age

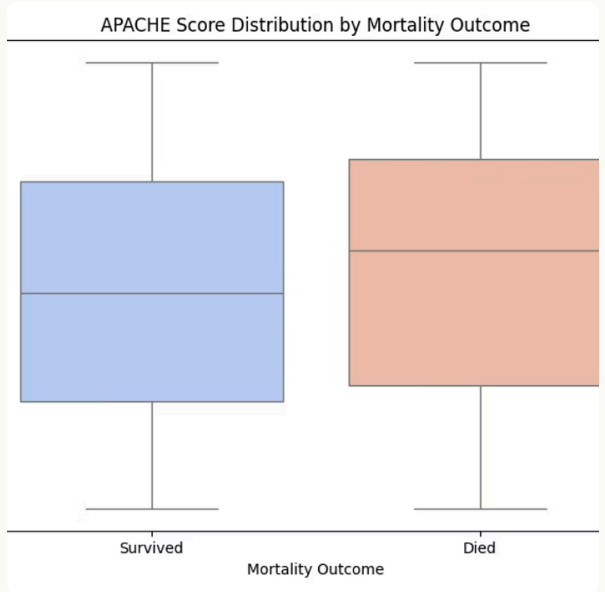
Exploratory Data Analysis (EDA)

Before training, we explored the dataset to understand data quality, feature behavior, and modeling challenges.



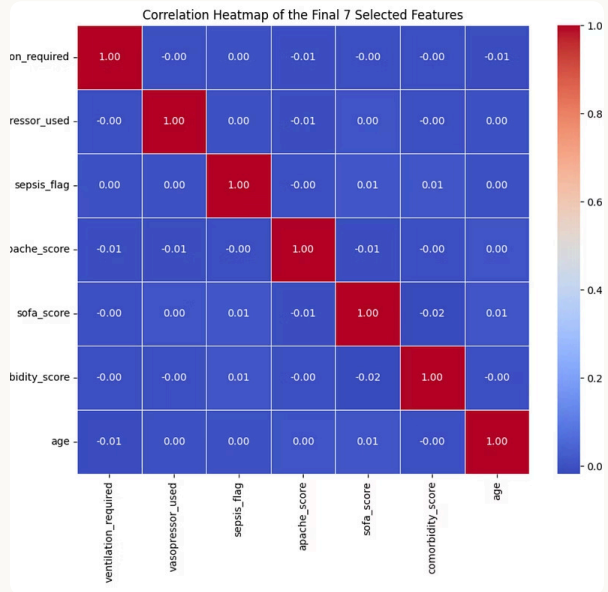
Lactate Levels on Patients (Histogram)

- Histograms showed the spread of numerical clinical variables.
- Example: lactate levels were inspected to understand patient measurement patterns.



APACHE Score Distribution by Mortality Outcome (BoxPlot)

- Boxplots were used to detect unusual values and compare feature behavior between survived and died patients.
- Example: APACHE score tended to be higher for patients who died, showing its clinical relevance.



Correlation Heatmap of the Final 7 Selected Features (Heatmap)

- The heatmap showed that the final selected features were not strongly correlated with each other.
- This reduces redundancy and supports using them together in the model.

Key Findings

01

Mortality influenced by multiple interacting factors

02

Mortality cases are the minority class

03

APACHE score showed higher values among died patients

04

Lactate histogram helped inspect numerical feature distribution

05

Final selected features had low correlation, reducing redundancy

Impact: EDA guided feature refinement, imbalance handling, and recall-focused evaluation.

Experiment 1 & 2: From Baseline Modeling to Feature Refinement

EXPERIMENT 1

Initial Baseline Modeling

Purpose: Test how models perform using the initial selected feature set while the dataset was still imbalanced.

What we did:

- Used the initial **22-feature set**
- Trained and compared multiple baseline classifiers
- Applied **cross-validation** for reliable evaluation
- Evaluated using :
  - Accuracy, Precision, Recall, F1-score, and ROC-AUC
- Applied threshold tuning to improve mortality detection

Best model: Tuned XGBoost

Key insight: The model detected many mortality cases, but also produced many false positives. This showed that accuracy alone is not enough for ICU mortality prediction.

EXPERIMENT 2

Feature Refinement

Purpose: Improve model focus by reducing noisy or weak features and keeping clinically meaningful predictors.

What changed:

- Reduced features from **22 → 7**
- Used:
  - correlation analysis
  - statistical significance
  - clinical relevance
- Dataset was still imbalanced

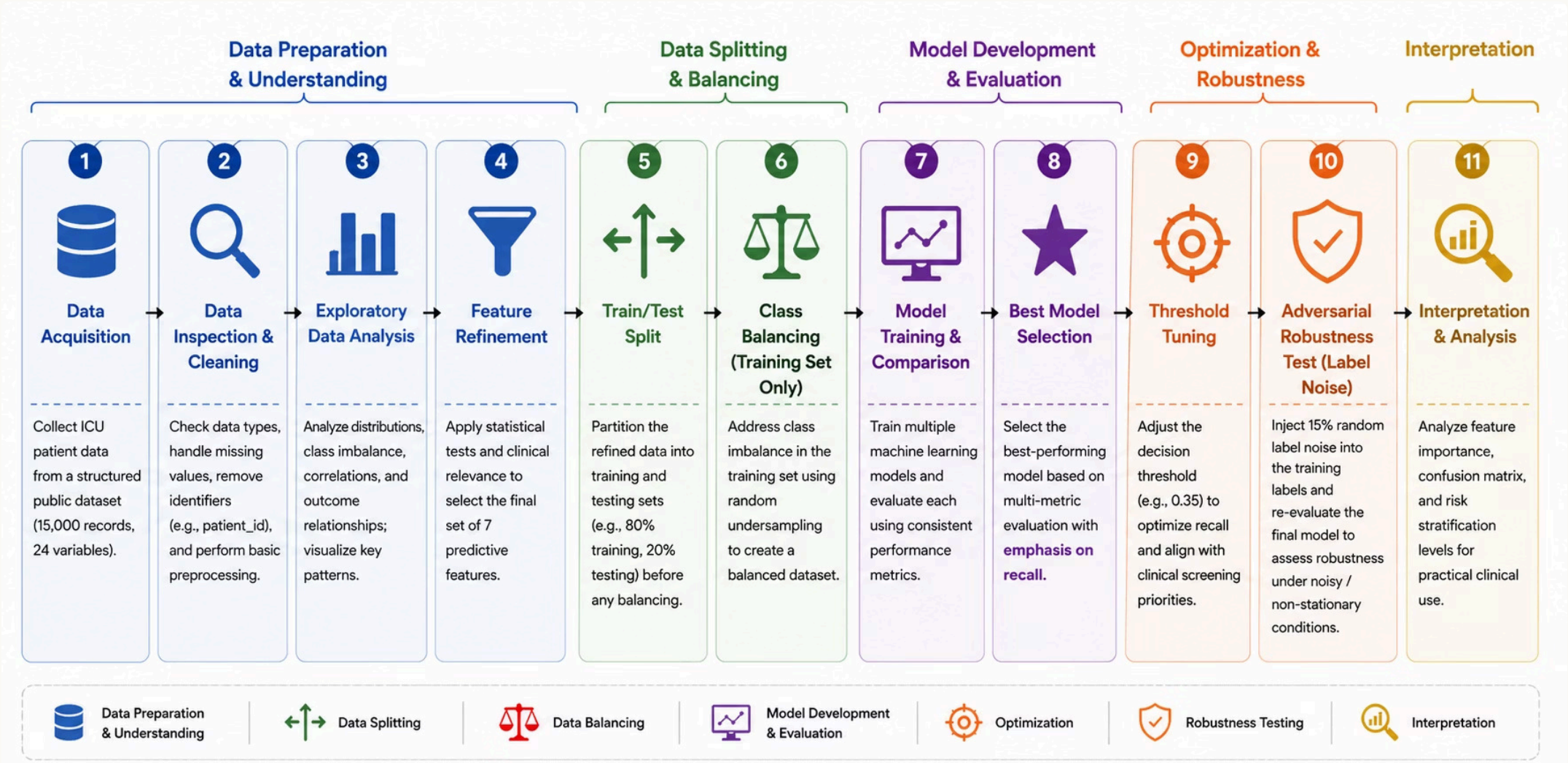
Final 7 selected features:

- ventilation_required
- vasopressor_used
- sepsis_flag
- apache_score
- sofa_score
- comorbidity_score
- age

> **Key insight:** Feature refinement improved model focus and interpretability, but imbalance still remained a major challenge.

Experiment 1 showed baseline limitations, and **Experiment 2** improved features — but class imbalance remained unresolved, leading to **Experiment 3**.

Methodology



Experiment 3: Final ML System — Balancing, Training & Optimization

CHALLENGE & IMBALANCE HANDLING

The Imbalance Challenge

After Experiment 2, we had our 7 final features, but the dataset remained severely imbalanced:

Class Distribution

Survived (0): 11,582 patients
Died (1): 3,418 patients

Why It Matters

Mortality was the minority class. This could bias the model towards survival, causing it to miss critical **high-risk ICU patients** (false negatives).

MODEL TRAINING & OPTIMIZATION

Extensive Model Training

We trained and compared **8 classifiers** on the balanced training data:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- KNN
- Naive Bayes
- Gradient Boosting
- XGBoost
- Model performance was validated using cross-validation to ensure reliable and generalizable results.

Evaluation Metrics

Accuracy, Precision, **Recall**, F1-score, ROC-AUC

Performance trade-offs were analyzed under both clean and noisy conditions.

Optimization Strategy

Focused on XGBoost and fine-tuned decision thresholds to optimize performance for mortality prediction:

- 0.50 (default)
- 0.35
- 0.40
- 0.45

Robustness Testing (Label Noise)

- Introduced **15% label noise** in training data
- Simulates real-world clinical data imperfections
- Result:
  - Recall ↑
  - Precision ↓
- Insight:
  - Noisy data increases false positives

Our Solution

- Treated as adversarial condition: **minority class suppression**.
- Balanced the **training set only** to prevent data leakage and maintain test set realism.
- Evaluated robustness under **label noise conditions**
- **Goal:** Enable the model to learn both survival and mortality patterns effectively.

Evaluation & Best Model Results

BEST MODEL RESULTS

Selected Model

Tuned XGBoost (0.35) : Selected based on best balance of recall, F1-score, and ROC-AUC.

58.0%
Accuracy

55.0%
Precision

88.45%
Recall

67.83%
F1-score

64.15%
ROC-AUC

Clinical Interpretation

This model was chosen for its high recall, minimizing missed high-risk patients. In ICU settings, a false alarm is less detrimental than failing to identify a patient at risk of mortality.

The model also demonstrated robustness under noisy data, though with a precision–recall trade-off.

Prototype Demonstration

Demonstrates how the final model can be used in an interactive decision-support interface

Inputs: clinical features → Outputs: mortality probability, class, and risk category

Google Docs

UI for ml & Datascience.MOV

Limitations, Future Work & Conclusion

LIMITATIONS

- Overall accuracy and ROC-AUC are still **moderate**.
- **High recall** improves detection of high-risk patients, but it also increases **false positives**.
- Class imbalance still affects performance.
- The dataset may not fully represent all real ICU patient populations.
- Not a real clinical system

FUTURE WORK

- Validate the model using **real hospital ICU data**.
- Add more clinical variables and **time-series** patient data.
- **Improve** model calibration and **robustness**.
- Use explainable AI (SHAP)
- Deploy in real-time systems

CONCLUSION

We developed a machine learning system for ICU mortality risk prediction. Applied: preprocessing, feature refinement, imbalance handling, threshold tuning, robustness testing.

Through data preprocessing, feature refinement, imbalance handling, model comparison, and threshold tuning, we selected **Tuned XGBoost (0.35)** as the final model which gave us high recall (88.45%), which supports **early identification of high-risk ICU patients**.

This project demonstrates how machine learning can support early risk detection in healthcare when clinical relevance, recall, and interpretability are prioritized.

Thank You

Any Questions?

Project: Mortality Risk Prediction using Machine Learning

Course: Machine Learning in healthcare

We welcome your feedback and questions